A novel audio watermarking in wavelet domain

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   - Adapt embedding strength and data embedding
   - Coefficients updating after data embedding
   - SNR estimation
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4 Summary of Main Results

Brian Chen and Gregory W. Wornell (2001)

They presented a class of information embedding methods called quantization index modulation (QIM) for digital watermarking.

Jiwu Huang, Yong Wang, and Yun Q. Shi (2002)

The watermark is placed in blockwise DCT coefficients while the synchronization signals are embedded in time domain to achieve self-synchronization.

Shaoquan Wu, Jiwu Huang, Daren Huang, and Yun Q. Shi (2005)

- Synchronization codes and watermarks are embedded together into low-frequency subband in DWT domain.

. . .	Sync-Code	Info-Bits	Sync-Code	Info-Bits	. . .

- The time-frequency localization capability of DWT is exploited to improve the efficiency in searching synchronization codes.

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$$c_i' = \begin{cases} \lfloor c_i/S \rfloor \cdot S + 3S/4 & \text{if } m_i = 1 \\ \lfloor c_i/S \rfloor \cdot S + S/4 & \text{if } m_i = 0 \end{cases}$$

watermark embedding

$$m_i^* = \begin{cases} 1 & \text{if } c_i^* - \lfloor c_i^*/S \rfloor \cdot S \geq S/2 \\ 0 & \text{if } c_i^* - \lfloor c_i^*/S \rfloor \cdot S < S/2 \end{cases}$$

watermark extraction

Shijun Xiang and Jiwu Huang (2006)

1. The possible degradation of audio watermarking caused by the DA/AD is modeled as following:
   - noise corruption
   - temporal scaling
   - amplitude modification
2. quantization-based audio watermarking is very susceptible to the DA/AD conversion.

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Problems

- The embedding strength of Wu's method is fixed, thus the secretly is not reliable against histogram steganalysis.

Go to histogram analysis

- Wu's method is susceptible to DA/AD due to the amplitude modification and noise corruption.

Solutions

- Employ AC energy to adjust embedding strength.
- The differences of blockwise low subband wavelet coefficients are used for Quantization.
- Adaptively control the embedding strength by audio's content.

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AC energy and quantization carrier

The AC energy is calculated as :

$$E_s = \sum_{i=1}^{N_w} (W_i - \overline{W})^2 \tag{1}$$

Then the total embedding strength in one section is generated:

$$\sum_{i=1}^{L} S_i = \sqrt{E_s} \cdot S_0 / S_T \tag{2}$$

The quantization carrier to embed the watermark bit:

$$C_i = \sum_{j=1}^{k/2} W_j^{l_i} - \sum_{j=k/2+1}^{k} W_j^{l_i} = \sum_{j=1}^{k/2} A_j^i - \sum_{j=1}^{k/2} B_j^i \tag{3}$$

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Adapt embedding strength and data embedding

1. Go to wu's method
2. The total embedding strength in every section have been dynamically adjusted by AC energy.
3. Embedding strength of every block in each section can be automatically modified.
4. The variation of embedding strength in each section must be automatically detected in the data extraction step.
5. An iteration is need. Go to detail iteration progress
6. C_i turns to be $C_i^{'}$. Go to an example

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Coefficients updating after data embedding

After data embedding, C_i turns to be $C_i^{'}$, thus the coefficients are updated as:

$$\begin{aligned} \{(A_j^i)'\} &= \{A_j^i\} + \tfrac{1}{2}(C_i' - C_i) \cdot (R_j / \textstyle\sum_{i=1}^{K/2} R_i) \\ & \qquad j = 1, \ldots, K/2 \\ \{(B_j^i)'\} &= \{B_j^i\} - \tfrac{1}{2}(C_i' - C_i) \cdot (R_j / \textstyle\sum_{i=1}^{K/2} R_i) \\ & \qquad j = 1, \ldots, K/2 \end{aligned} \tag{4}$$

R denotes a scale window sequence.

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SNR estimation

In Wu's method:

$$SNR = -10log_{10}(\frac{7NS^2}{2^k 48\|F\|_2^2}) \tag{5}$$

In proposed method:

$$SNR = -10log_{10}(\frac{7\overline{R}^2}{48KLS_T^2} \cdot \frac{E_s}{E} \cdot S_0^2) + \delta_{SNR} \tag{6}$$

Go to detail proof

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The fluctuation of $\sqrt{E_s}$ in common signal processing manipulations.

Attacks	Max(%)	Min(%)	Average(%)
$GNC(15dB)$	1.01	5.25×10^{-7}	9.98×10^{-2}
$Resampling(6kHz)$	1.90	8.54×10^{-6}	6.08×10^{-2}
$Requantization(8Bit)$	8.45	1.55×10^{-6}	8.04×10^{-2}
$MP3(32kbps)$	4.15	3.85×10^{-4}	0.37
$D/A - A/Dconversion$	9.27	8.27×10^{-5}	0.31

The fluctuation of $\sqrt{E_s}$ by watermark embedding processing.

$SNR(dB)$	Max(%)	Min(%)	Average(%)
20	10.18	8.29×10^{-5}	0.39
25	2.82	7.73×10^{-5}	0.13
30	0.63	1.60×10^{-5}	5.09×10^{-2}
35	0.27	3.31×10^{-5}	2.68×10^{-2}
40	0.17	6.38×10^{-6}	1.43×10^{-2}

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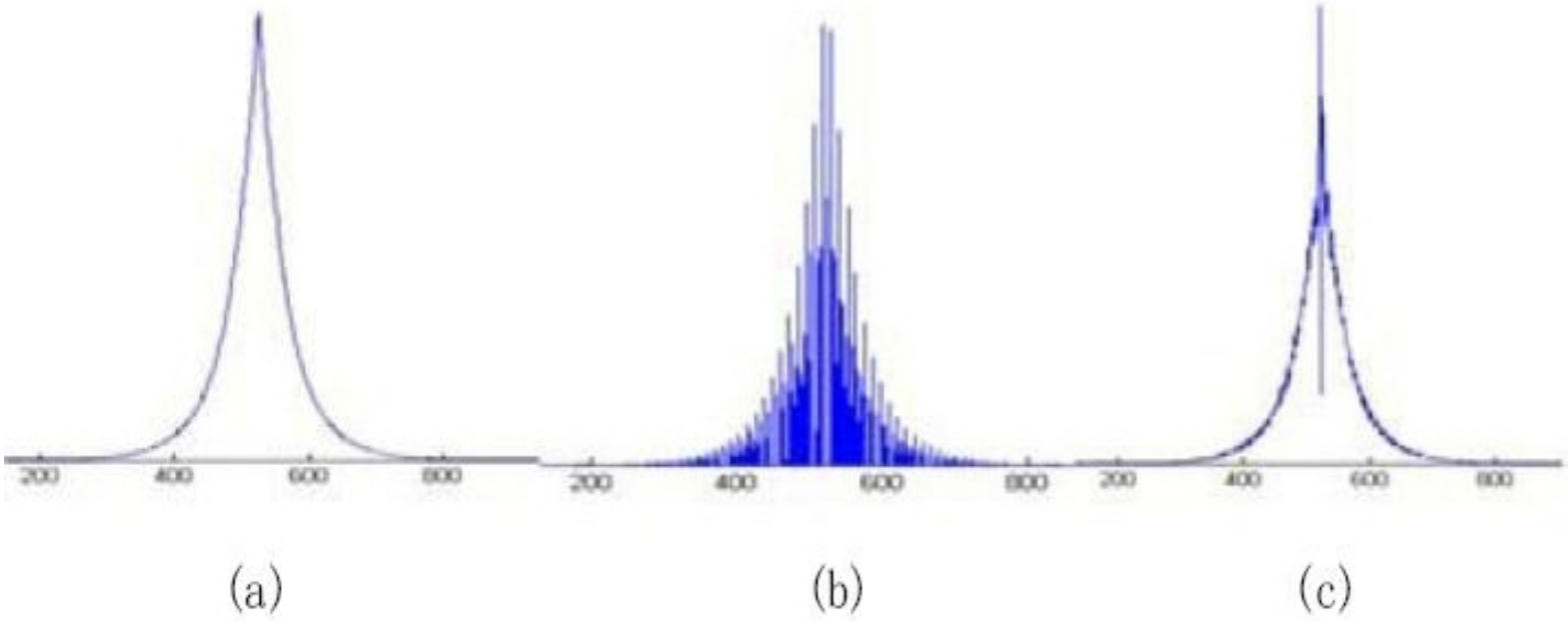
Histogram Analysis



Histogram of wavelet coefficient

(a) clean audio; (b) Wu's method (steady embedding strength); (c) proposed method (adaptive embedding strength)

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Subjective Diff-Grades and Average SDG score

SDG	Description
0.0	Imperceptible
-1.0	Perceptible, but not annoying
-2.0	Slightly annoying
-3.0	Annoying
-4.0	Very annoying

	Piano	Rock	E-guitar	Organ	Pop
Average score of SDG	-0.2	-0.1	0	-0.2	0

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Robustness

	Piano	Rock	E-guitar	Organ	Pop
Gaussian noise (14dB)	8.78	0.19	0.02	0.58	0.02
Gaussian noise (20dB)	2.31	0.06	0.00	0.02	0.00
Resampling (6000Hz)	0.00	0.11	0.02	0.00	0.06
Requantization (8bits)	0.80	0.08	0.00	0.00	0.00
MP3 (128kbps)	0.00	0.00	0.00	0.00	0.00
MP3 (56kbps)	0.00	0.39	0.14	0.00	0.02
MP3 (32kbps)	0.08	1.95	0.25	0.25	0.30
D/A-A/D conversion	0.00	0.31	0.03	3.08	0.78

The BER(%) of manipulations.

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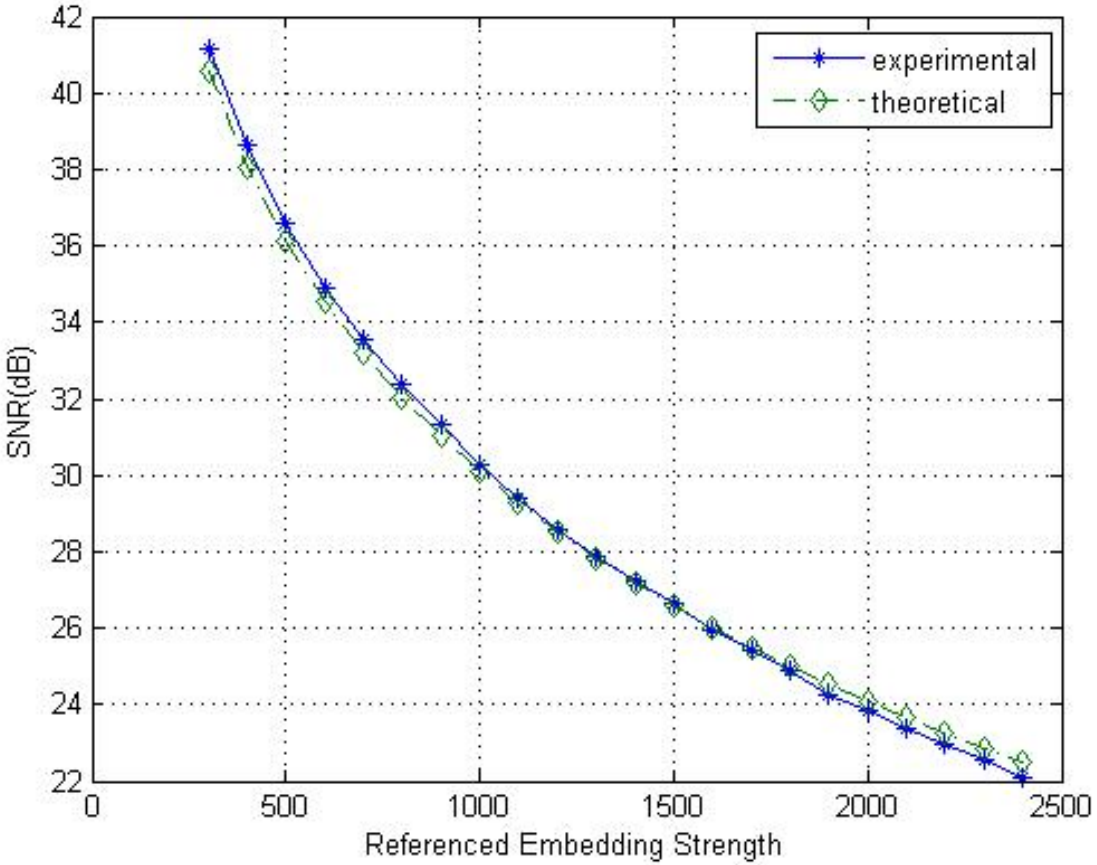

SNR and Referenced Embedding Strength S_0 for *pop.wav*.

Summary

1. make the security more reliable by adaptively control the embedding strength
2. make the watermark more robust for AD/DA conversion by use the difference of consecutive blockwise coefficients
3. provide accurate SNR estimation for watermark embedding

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Thanks!

- Step 1. Arbitrarily initialize $Y_i^{(0)}$ as a uniform array.
- Step 2. Update $\{S_i\}$ and $\{Y_i\}$ by the flow as follows (7,8,9):

$$S_i^{(n)} = \begin{cases} \overline{S} & \text{if } T \geq \overline{S} \\ \overline{S}/2 & \text{if } T \leq \overline{S}/2 \\ T & \overline{S}/2 < T < \overline{S} \end{cases}$$

$$\text{where} \quad T = G \cdot \frac{\frac{1}{(|Y_i^{n-1}|)}}{\sum_{i=1}^{L} \frac{1}{(|Y_i^{n-1}|)}} \tag{7}$$

$$S_i^{(n)} = (S_i^{(n)} + S_i^{(n-1)})/2 \qquad \text{if } (n \geq 2) \tag{8}$$

$$Y_i^{(n)} = \lfloor |C_i^{(n)}|/S_i^{(n)} \rfloor \cdot S_i^{(n)} + S_i^{(n)} \cdot (1/2 + 1/4 \cdot m_i) \tag{9}$$

- Step 3. Compute the distance between $Y_i^{(n)}$ and $Y_i^{(n-1)}$:

$$\xi^{(n)} = Y_i^{(n)} - Y_i^{(n-1)} \tag{10}$$

- Step 4. If $\xi^{(n)} > \xi_0$, the iteration is not convergent, then go to Step 2; else if $\xi^{(n)} < \xi_0$, the iteration is convergent, go to Step 5.
- Step 5. Terminate the iteration, and $\{S_i^{(n+1)}\}$ is as the same as $\{S_i^{(n)}\}$.
- Step 6. Let $\{Y_i\}$ be the new $\{C_i'\}$.

Return

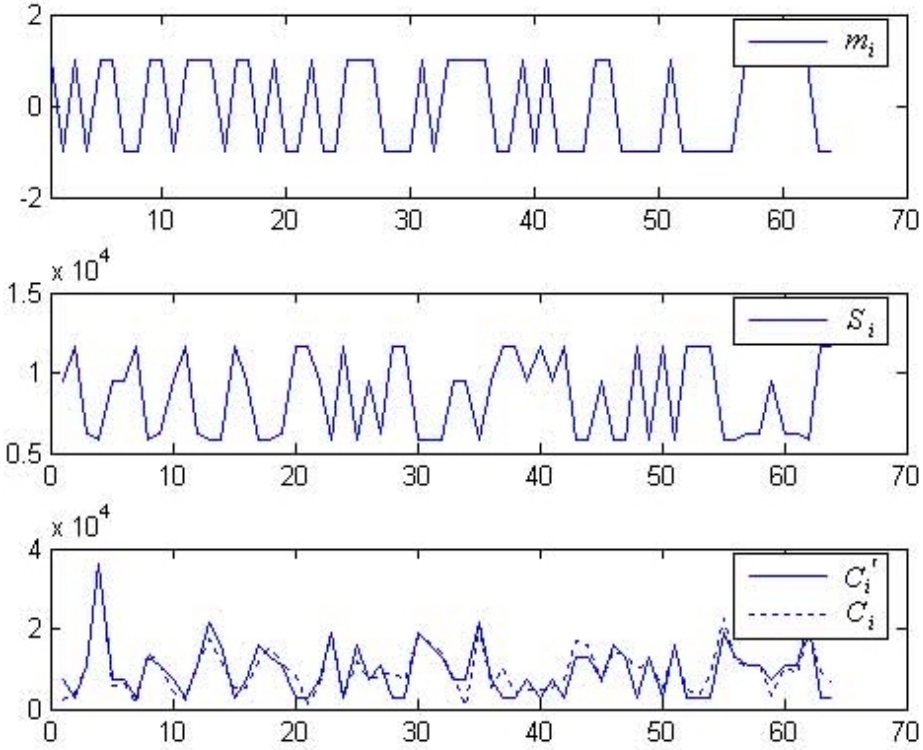
Return

$$\overline{S_w} = \frac{\overline{R}}{K \cdot L}\sqrt{E_s} \cdot S_0 / S_T \tag{11}$$

$$\overline{R} = \sqrt{\frac{1}{K/2}\sum_{j=1}^{K/2}\left(\frac{K}{2} \cdot \frac{R_j}{\sum_{i=1}^{K/2} R_i}\right)^2} \tag{12}$$

From Wu's analysis, the difference of a section in wavelet domain between clean and watermarked audio is:

$$Exp.((W_i - W_i')^2) = \frac{7\overline{S_w}^2 KL}{48} = \frac{7\overline{R}^2}{48KL} E_s \cdot S_0^2 / S_T^2 \tag{13}$$



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